ASSIGNMENT 5.1 & 6

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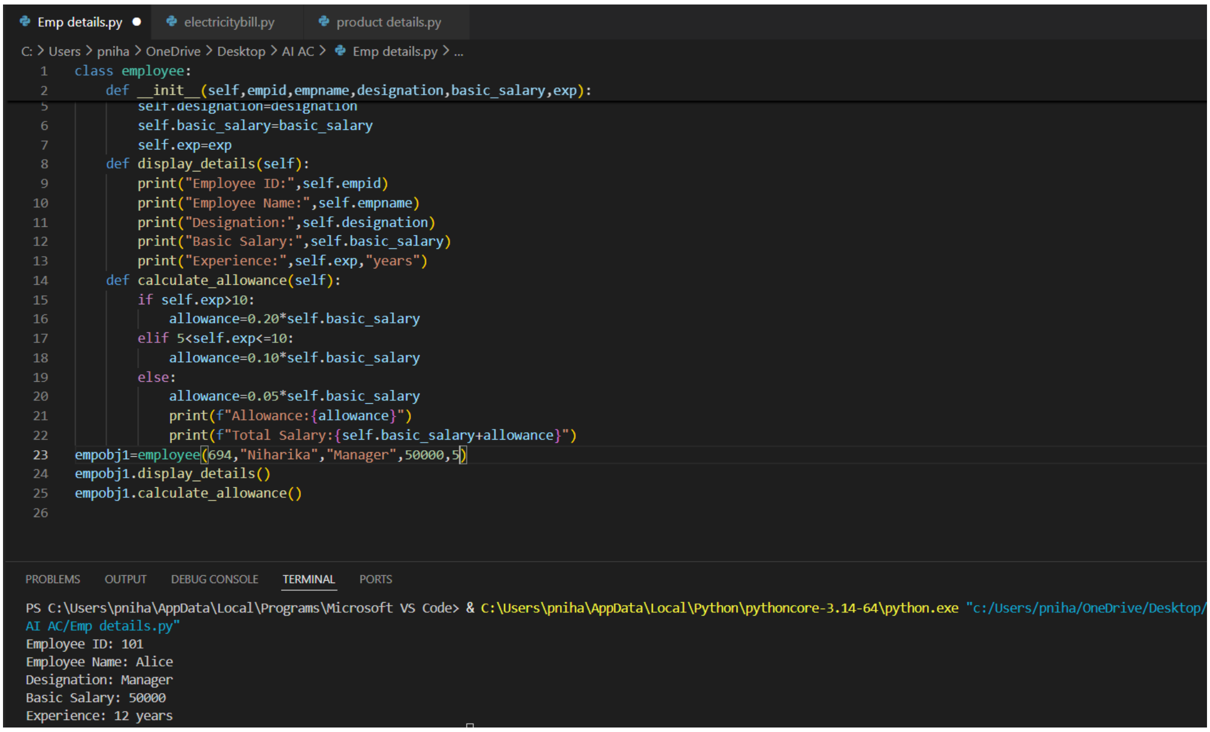
Batch: 24

Task 1:

Employee Data: Create Python code that defines a class named `Employee` with the following attributes: `empid`, `empname`, `designation`, `basic_salary`, and `exp`. Implement a method `display_details()` to print all employee details. Implement another method `calculate_allowance()` to determine additional allowance based on experience:

- If `exp > 10 years` → allowance = 20% of `basic_salary`
- If `5 ≤ exp ≤ 10 years` → allowance = 10% of `basic_salary`
- If `exp < 5 years` → allowance = 5% of `basic_salary`

Finally, create at least one instance of the `Employee` class, call the `display_details()` method, and print the calculated allowance.



```
Emp details.py    electricitybill.py    product details.py
C: > Users > pniha > OneDrive > Desktop > AI AC > Emp details.py > ...
class employee:
    def __init__(self,empid,empname,designation,basic_salary,exp):
        self.designation=designation
        self.basic_salary=basic_salary
        self.exp=exp
    def display_details(self):
        print("Employee ID:",self.empid)
        print("Employee Name:",self.empname)
        print("Designation:",self.designation)
        print("Basic Salary:",self.basic_salary)
        print("Experience:",self.exp,"years")
    def calculate_allowance(self):
        if self.exp>10:
            allowance=0.20*self.basic_salary
        elif 5<self.exp<=10:
            allowance=0.10*self.basic_salary
        else:
            allowance=0.05*self.basic_salary
            print(f"Allowance:{allowance}")
            print(f"Total Salary:{self.basic_salary+allowance}")
empobj1=employee(694,"Niharika","Manager",50000,5)
empobj1.display_details()
empobj1.calculate_allowance()

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/pniha/OneDrive/Desktop/
AI AC/Emp details.py"
Employee ID: 101
Employee Name: Alice
Designation: Manager
Basic Salary: 50000
Experience: 12 years
```

Task 2:

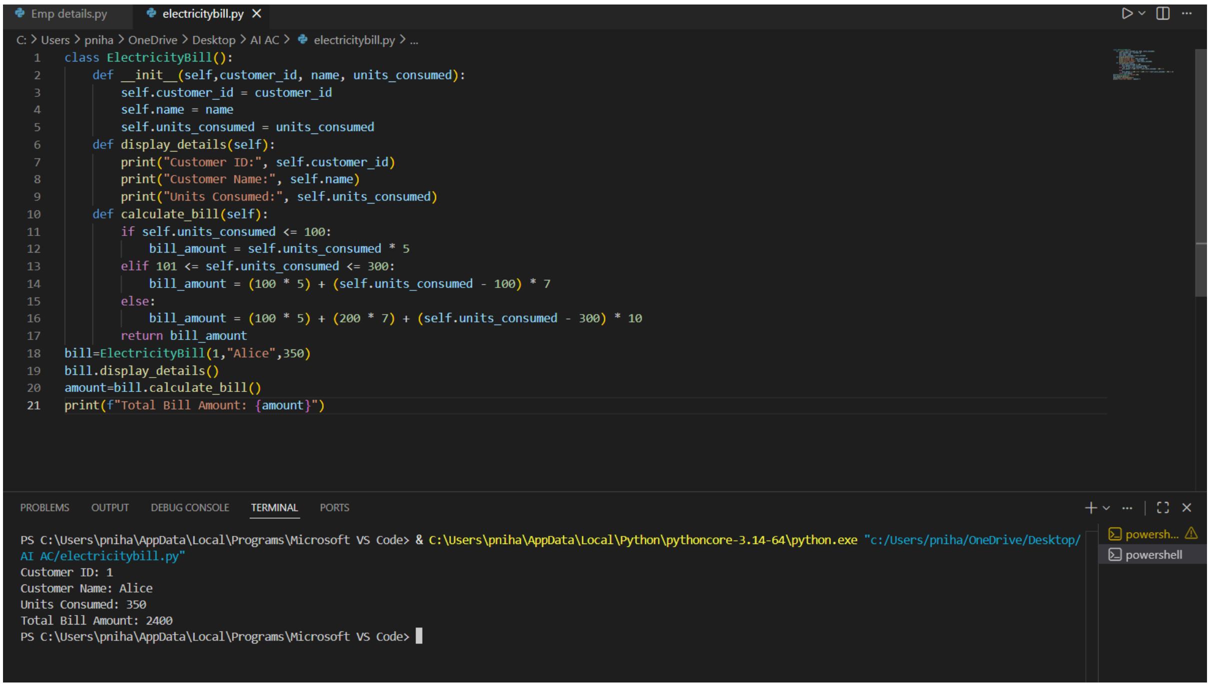
Electricity Bill Calculation- Create Python code that defines a class named `ElectricityBill` with attributes: `customer_id`, `name`, and `units_consumed`. Implement a method `display_details()` to print customer details, and a method `calculate_bill()` where:

- Units ≤ 100 → ₹5 per unit

- 101 to 300 units → ₹7 per unit

- More than 300 units → ₹10 per unit

Create a bill object, display details, and print the total bill amount.



```python
class ElectricityBill():
    def __init__(self,customer_id, name, units_consumed):
        self.customer_id = customer_id
        self.name = name
        self.units_consumed = units_consumed
    def display_details(self):
        print("Customer ID:", self.customer_id)
        print("Customer Name:", self.name)
        print("Units Consumed:", self.units_consumed)
    def calculate_bill(self):
        if self.units_consumed <= 100:
            bill_amount = self.units_consumed * 5
        elif 101 <= self.units_consumed <= 300:
            bill_amount = (100 * 5) + (self.units_consumed - 100) * 7
        else:
            bill_amount = (100 * 5) + (200 * 7) + (self.units_consumed - 300) * 10
        return bill_amount
bill=ElectricityBill(1,"Alice",350)
bill.display_details()
amount=bill.calculate_bill()
print(f"Total Bill Amount: {amount}")
```

```
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/pniha/OneDrive/Desktop/AI AC/electricitybill.py"
Customer ID: 1
Customer Name: Alice
Units Consumed: 350
Total Bill Amount: 2400
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code>
```

Task 3:

Product Discount Calculation- Create Python code that defines a class named `Product` with attributes: `product_id`, `product_name`, `price`, and `category`. Implement a method `display_details()` to print product details. Implement another method `calculate_discount()` where:

- Electronics → 10% discount

- Clothing → 15% discount

- Grocery → 5% discount

Create at least one product object, display details, and print the final price after discount.

```python
class Product:
    def __init__(self, product_id, product_name, price, category):
        self.product_id=product_id
        self.product_name=product_name
        self.price=price
        self.category=category
    def display_details(self):
        print(f"Product ID:{self.product_id}")
        print(f"Product Name:{self.product_name}")
        print(f"Price:{self.price}")
        print(f"Category:{self.category}")
    def calculate_discount(self):
        if self.category.lower() == "electronics":
            discount = 0.10 * self.price
        elif self.category.lower() == "clothing":
            discount = 0.15 * self.price
        else:
            discount = 0.05 * self.price
        print(f"Discount Amount: {discount}")
        print(f"Price after Discount: {self.price - discount}")
prod1=Product(301,"Smartwatch",50000,"Electronics")
prod1.display_details()
prod1.calculate_discount()
```

```
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/pniha/OneDrive/Desktop/AI AC/product details.py"
Product ID:301
Product Name:Smartwatch
Price:50000
Category:Electronics
Discount Amount: 5000.0
Price after Discount: 45000.0
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code>
```

Task 4:

Book Late Fee Calculation- Create Python code that defines a class named `LibraryBook` with attributes: `book_id`, `title`, `author`, `borrower`, and `days_late`. Implement a method `display_details()` to print book details, and a method `calculate_late_fee()` where:

- Days late ≤ 5 → ₹5 per day

- 6 to 10 days late → ₹7 per day

- More than 10 days late → ₹10 per day

Create a book object, display details, and print the late fee.

```python
class librarybook:
    def __init__(self,book_id,title,author,borrower,days_late):
        self.book_id = book_id
        self.title = title
        self.author = author
        self.borrower = borrower
        self.days_late = days_late
    def display_details(self):
        print(f"Book ID: {self.book_id}")
        print(f"Title: {self.title}")
        print(f"Author: {self.author}")
        print(f"Borrower: {self.borrower}")
        print(f"Days Late: {self.days_late}")
    def calculate_late_fee(self):
        if self.days_late <= 5:
            late_fee=self.days_late * 5
        elif 6 <= self.days_late <= 10:
            late_fee = self.days_late * 7
        else:
            late_fee = self.days_late * 10
        print(f"Late Fee: {late_fee}")
librarybook1 = librarybook("B015", "The Alchemist", "Paluo Coeho", "Niharika", 2)
librarybook1.display_details()
librarybook1.calculate_late_fee()
```

```
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/pniha/On
AI AC/booklate.py"
Book ID: B015
Title: The Alchemist
Author: Paluo Coeho
Borrower: Niharika
Days Late: 2
Late Fee: 10
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code>
```

Task 5:

Student Performance Report - Define a function `student_report(student_data)` that accepts a dictionary containing student names and their marks. The function should:

- Calculate the average score for each student
- Determine pass/fail status (pass ≥ 40)
- Return a summary report as a list of dictionaries

Use Copilot suggestions as you build the function and format the output.

Task 6:

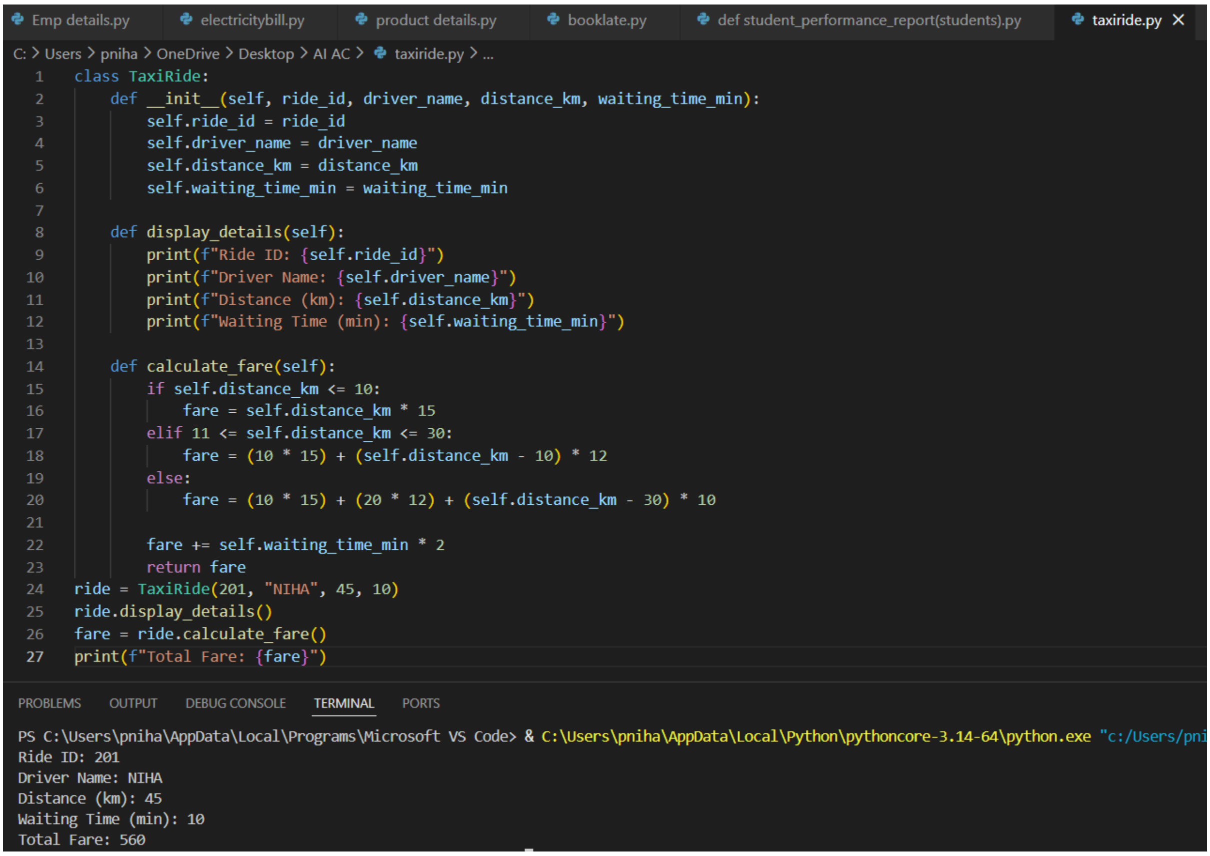
Taxi Fare Calculation-Create Python code that defines a class named `TaxiRide` with attributes: `ride_id`, `driver_name`, `distance_km`, and `waiting_time_min`. Implement a method `display_details()` to print ride details, and a method `calculate_fare()` where:

- ₹15 per km for the first 10 km
- ₹12 per km for the next 20 km

- ₹10 per km above 30 km

- Waiting charge: ₹2 per minute

Create a ride object, display details, and print the total fare.



Emp details.py | electricitybill.py | product details.py | booklate.py | def student_performance_report(students).py | taxiride.py

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```python
class TaxiRide:
    def __init__(self, ride_id, driver_name, distance_km, waiting_time_min):
        self.ride_id = ride_id
        self.driver_name = driver_name
        self.distance_km = distance_km
        self.waiting_time_min = waiting_time_min

    def display_details(self):
        print(f"Ride ID: {self.ride_id}")
        print(f"Driver Name: {self.driver_name}")
        print(f"Distance (km): {self.distance_km}")
        print(f"Waiting Time (min): {self.waiting_time_min}")

    def calculate_fare(self):
        if self.distance_km <= 10:
            fare = self.distance_km * 15
        elif 11 <= self.distance_km <= 30:
            fare = (10 * 15) + (self.distance_km - 10) * 12
        else:
            fare = (10 * 15) + (20 * 12) + (self.distance_km - 30) * 10

        fare += self.waiting_time_min * 2
        return fare
ride = TaxiRide(201, "NIHA", 45, 10)
ride.display_details()
fare = ride.calculate_fare()
print(f"Total Fare: {fare}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pniha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/pni
Ride ID: 201
Driver Name: NIHA
Distance (km): 45
Waiting Time (min): 10
Total Fare: 560
```

Task 7:

Statistics Subject Performance - Create a Python function

`statistics_subject(scores_list)` that accepts a list of 60 student scores

and computes key performance statistics. The function should return

the following:

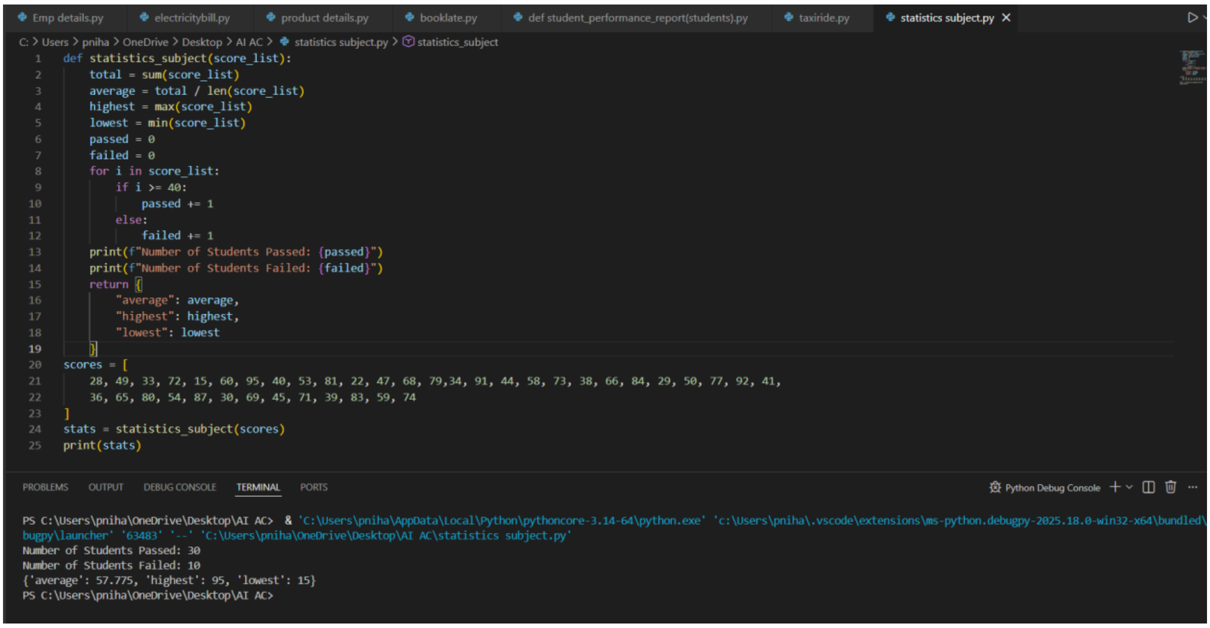
- Highest score in the class

- Lowest score in the class

- Class average score

- Number of students passed (score ≥ 40)

- Number of students failed (score < 40)



```python
def statistics_subject(score_list):
    total = sum(score_list)
    average = total / len(score_list)
    highest = max(score_list)
    lowest = min(score_list)
    passed = 0
    failed = 0
    for i in score_list:
        if i >= 40:
            passed += 1
        else:
            failed += 1
    print(f"Number of Students Passed: {passed}")
    print(f"Number of Students Failed: {failed}")
    return {
        "average": average,
        "highest": highest,
        "lowest": lowest
    }
scores = [
    28, 49, 33, 72, 15, 60, 95, 40, 53, 81, 22, 47, 68, 79,34, 91, 44, 58, 73, 38, 66, 84, 29, 50, 77, 92, 41,
    36, 65, 80, 54, 87, 30, 69, 45, 71, 39, 83, 59, 74
]
stats = statistics_subject(scores)
print(stats)
```

```
PS C:\Users\pniha\OneDrive\Desktop\AI AC>  & 'C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\pniha\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\bugpy\launcher' '63483' '--' 'C:\Users\pniha\OneDrive\Desktop\AI AC\statistics subject.py'
Number of Students Passed: 30
Number of Students Failed: 10
{'average': 57.775, 'highest': 95, 'lowest': 15}
PS C:\Users\pniha\OneDrive\Desktop\AI AC>
```

Task Description #8 (Transparency in Algorithm Optimization)

Task: Use AI to generate two solutions for checking prime numbers:

- Naive approach(basic)
- Optimized approach

Prompt:

“Generate Python code for two prime-checking methods and explain

how the optimized version improves performance.”

Expected Output:

- Code for both methods.
- Transparent explanation of time complexity.
- Comparison highlighting efficiency improvements.

Emp details.py | electricitybill.py | product details.py | booklate.py | def student_performance_report(stude

C: > Users > pniha > OneDrive > Desktop > AI AC > # that reads a file a.py > read_file

```python
# generate a program that reads a file and process the data
# Generate code with proper error handling and clear explanations for each exception.

def read_file(file_path):
    try:
        # Attempt to open the file
        with open(file_path, 'r') as file:
            data = file.read()
            print("File content successfully read.")
            return data
    except FileNotFoundError:
        # Handle the case where the file does not exist
        print(f"Error: The file at {file_path} was not found.")
    except PermissionError:
        # Handle the case where there are permission issues
        print(f"Error: You do not have permission to read the file at {file_path}.")
    except Exception as e:
        # Handle any other exceptions that may occur
        print(f"An unexpected error occurred: {e}")
file_path = 'example.txt'   # Specify the path to your file here
file_content = read_file(file_path)
if file_content:
    print("File Content:")
    print(file_content)
```

```
File content successfully read.
File Content:
Hello Everyone
Welcome to AI Assisted Coding class
Third year second semester
SR University
Lets work with files as part of lab assignment
```

Task Description #9 (Transparency in Recursive Algorithms)

Objective: Use AI to generate a recursive function to calculate

Fibonacci numbers.

Instructions:

1. Ask AI to add clear comments explaining recursion.

2. Ask AI to explain base cases and recursive calls.

Expected Output:

• Well-commented recursive code.

• Clear explanation of how recursion works.

• Verification that explanation matches actual execution.

```python
# write a code to generate a recursive function to calculate fibonacci numbers.
# - add clear comments explaining recusrion.
# - also explain base cases and recursive calls.
# - verification that explanation matches actual execution.
def fibonacci(n):
    """
    Calculate the nth Fibonacci number using recursion.

    The Fibonacci sequence is defined as:
    F(0) = 0 (base case)
    F(1) = 1 (base case)
    F(n) = F(n-1) + F(n-2) for n > 1 (recursive case)

    Parameters:
    n (int): The position in the Fibonacci sequence to calculate.

    Returns:
    int: The nth Fibonacci number.
    """
    # Base cases
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        # Recursive case: sum of the two preceding numbers
        return fibonacci(n - 1) + fibonacci(n - 2)
# Example usage and verification
n = 6
print(f"The {n}th Fibonacci number is: {fibonacci(n)}")
# Explanation:
# When we call fibonacci(6), the function checks if n is 0 or 1. Since it's neither, it proceeds to the recursive case:
# fibonacci(6) = fibonacci(5) + fibonacci(4)
# This pattern continues, breaking down each call until it reaches the base cases:
# fibonacci(1) = 1 and fibonacci(0) = 0.
# The results are then combined back up the call stack to produce the final result.
# The execution for fibonacci(6) would look like this:
# fibonacci(6)
# = fibonacci(5) + fibonacci(4)
# = (fibonacci(4) + fibonacci(3)) + (fibonacci(3) + fibonacci(2))
# = ((fibonacci(3) + fibonacci(2)) + (fibonacci(2) + fibonacci(1))) + ((fibonacci(2) + fibonacci(1)) + (fibonacci(1) + fibonacci(0)))
# = ... and so on, until all calls reach the base cases.
# The final result is 8, which is the 6th Fibonacci number.
```

```
PS C:\Users\pniha\OneDrive\Pictures\Screenshots>  & 'C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\pniha\.vscode\extensions\ms-python.debugpy-2025.18
8' '--' 'C:\Users\pniha\OneDrive\Pictures\Screenshots\fibonacci numbers.py'
The 6th Fibonacci number is: 8
PS C:\Users\pniha\OneDrive\Pictures\Screenshots>
```

Task Description #10 (Transparency in Error Handling)

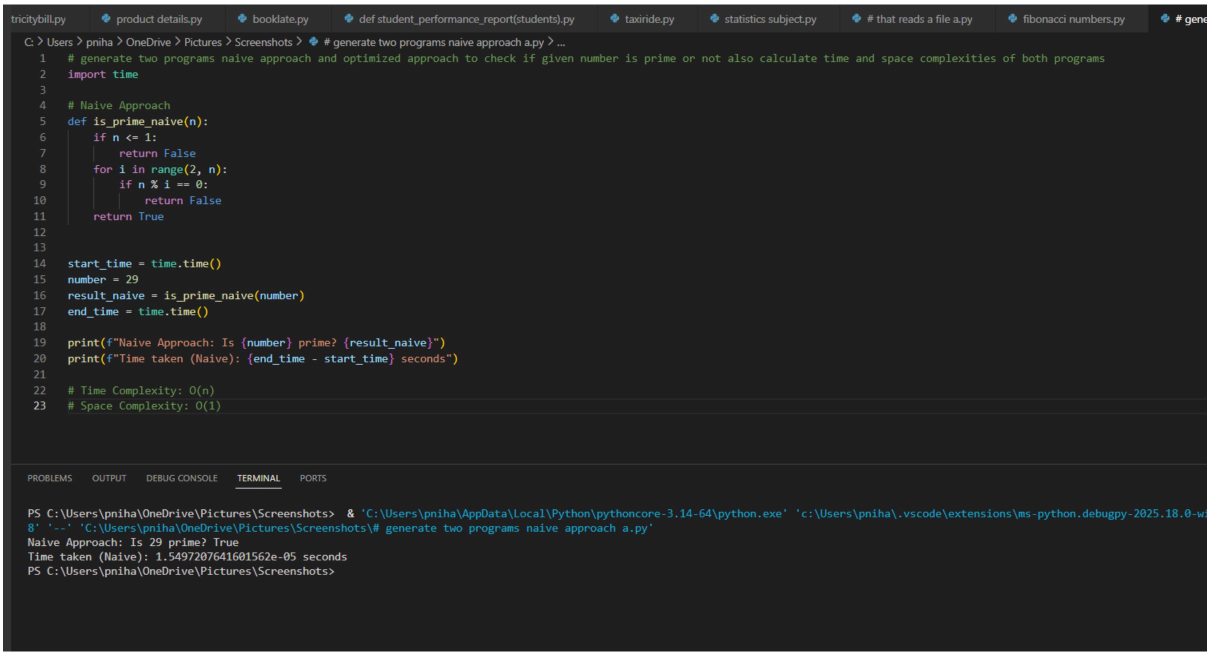
Task: Use AI to generate a Python program that reads a file and processes data.

Prompt:

“Generate code with proper error handling and clear explanations for each exception.”

Expected Output:

- Code with meaningful exception handling.
- Clear comments explaining each error scenario.
- Validation that explanations align with runtime behavior.



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product details.py
booklate.py
def student_performance_report(students).py
taxiride.py
statistics subject.py
that reads a file a.py
fibonacci numbers.py
C: > Users > pniha > OneDrive > Pictures > Screenshots > # generate two programs naive approach a.py > ...
# generate two programs naive approach and optimized approach to check if given number is prime or not also calculate time and space complexities of both programs
import time
# Naive Approach
def is_prime_naive(n):
if n <= 1:
return False
for i in range(2, n):
if n % i == 0:
return False
return True
start_time = time.time()
number = 29
result_naive = is_prime_naive(number)
end_time = time.time()
print(f"Naive Approach: Is {number} prime? {result_naive}")
print(f"Time taken (Naive): {end_time - start_time} seconds")
# Time Complexity: O(n)
# Space Complexity: O(1)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\pniha\OneDrive\Pictures\Screenshots> & 'C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\pniha\.vscode\extensions\ms-python.debugpy-2025.18.0-w
8' '--' 'C:\Users\pniha\OneDrive\Pictures\Screenshots\# generate two programs naive approach a.py'
Naive Approach: Is 29 prime? True
Time taken (Naive): 1.5497207641601562e-05 seconds
PS C:\Users\pniha\OneDrive\Pictures\Screenshots>